

Finite-Loop Point-Defect Transport in Irradiated α -Zirconium

A Unified Continuum–Atomistic Formulation Connecting Woo’s Elastic Bias Theory and Atomistic Thermal-Drift Capture

Abstract

A finite dislocation loop in irradiated α -zirconium is not merely a geometrical sink. Its elastic field changes both the equilibrium free energy of a mobile point defect and the migration barrier through the saddle-point configuration. Classical continuum treatments usually retain only an isotropic defect–loop interaction and consequently predict nearly the same bias for vacancy- and interstitial-type loops of equal size. Woo showed that anisotropy of the point-defect elastic dipole tensor at the migration saddle point breaks this symmetry and produces an intrinsic bias differential between vacancy and interstitial loops. More recently, March-Rico and Wirth obtained loop-character-dependent capture directly from atomistic binding-energy maps in α -Zr and showed that thermal-drift capture can favor same-type defects.

The formulation developed here combines these ideas in a finite-loop transport problem. The loop is represented explicitly in three dimensions; its strain field is obtained from anisotropic HCP elasticity or a dislocation-dynamics solver; equilibrium and saddle-point defect energies are represented by elastic dipole tensors, atomistic energy maps, or a matched hybrid of both; local jump rates are constructed from transition-state theory; and the resulting position-dependent diffusion tensor is coupled to a drift–diffusion equation. The total point-defect flux into the finite loop directly defines the capture rate, sink strength, bias factor, and loop growth rate. Finite-element, finite-volume, kinetic Monte Carlo, and multiscale implementation strategies are described, together with verification tests and a practical computational workflow for α -Zr.

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1 Physical objective

The objective is to calculate, without prescribing an empirical bias factor, the four finite-loop capture rates

$$K_{i,a_I}, \quad K_{v,a_I}, \quad K_{i,a_V}, \quad K_{v,a_V}. \quad (1)$$

Using Woo’s convention,

$$B_{a_I} = 1 - \frac{Z_{v,a_I}}{Z_{i,a_I}}, \quad B_{a_V} = 1 - \frac{Z_{v,a_V}}{Z_{i,a_V}}. \quad (2)$$

The principal question is whether the experimentally relevant regime

$$\boxed{B_{a_I} > 0, \quad B_{a_V} < 0} \quad (3)$$

emerges from the coupled effects of finite-loop elasticity, HCP diffusion anisotropy, equilibrium defect–loop binding, migration saddle-point anisotropy, near-core atomistic reconstruction, and actual loop geometry. Woo identified this as a possible intrinsic-bias regime and discussed zirconium as a candidate material [2]; March-Rico and Wirth later obtained the same qualitative “same-type” tendency from atomistically calculated thermal-drift capture radii in α -Zr [7].

2 Finite-loop geometry and crystallography

2.1 Circular loop

For a circular loop of radius R_L , center $\mathbf{x}_0$, and orthonormal vectors $\mathbf{e}_1, \mathbf{e}_2$ spanning the loop plane,

$$\mathbf{n}_L = \mathbf{e}_1 \times \mathbf{e}_2, \quad (4)$$

and

$$\Gamma_L : \quad \mathbf{x}_L(\phi) = \mathbf{x}_0 + R_L(\mathbf{e}_1 \cos \phi + \mathbf{e}_2 \sin \phi), \quad 0 \leq \phi < 2\pi. \quad (5)$$

The local line tangent is

$$\boldsymbol{\xi}(\phi) = \frac{\partial \mathbf{x}_L / \partial \phi}{|\partial \mathbf{x}_L / \partial \phi|}. \quad (6)$$

For irradiation-induced a -loops in α -Zr,

$$\mathbf{b}_a = \frac{1}{3}\langle 11\bar{2}0 \rangle, \quad |\mathbf{b}_a| \simeq a, \quad (7)$$

and the dominant habit-plane family is prismatic, commonly near $\{10\bar{1}0\}$. The implementation should permit $\mathbf{n}_L$ to be specified independently of $\mathbf{b}_L$ so that first-order prism, second-order prism, and tilted habit planes can be represented.

2.2 Signed loop character

Define

$$s_L = \begin{cases} +1, & \text{interstitial loop,} \\ -1, & \text{vacancy loop,} \end{cases} \quad (8)$$

with a consistent loop-normal convention, and write

$$\mathbf{b}_L = s_L \mathbf{b}_0. \quad (9)$$

For geometrically identical loops of opposite sign in linear elasticity,

$$\boldsymbol{\varepsilon}^{-L}(\mathbf{x}) = -\boldsymbol{\varepsilon}^{+L}(\mathbf{x}), \quad \boldsymbol{\sigma}^{-L}(\mathbf{x}) = -\boldsymbol{\sigma}^{+L}(\mathbf{x}). \quad (10)$$

2.3 Core reaction surface

Surround the dislocation line by a small reaction tube

$$\Gamma_c = \{\mathbf{x} : d(\mathbf{x}, \Gamma_L) = r_c\}, \quad (11)$$

where d is shortest distance to the loop line. A species- and orientation-dependent r_c can be used, but in the most fundamental calculation r_c should remain a small core boundary and the long-range bias should emerge from the transport equations.

3 Anisotropic elastic field of the finite loop

For α -Zr, use the full hexagonal stiffness tensor

$$\sigma_{ij} = C_{ijkl}\varepsilon_{kl}^e, \quad (12)$$

with independent constants $C_{11}, C_{12}, C_{13}, C_{33}, C_{44}$ and $C_{66} = (C_{11} - C_{12})/2$. Away from the core,

$$\partial_j \sigma_{ij} = 0. \quad (13)$$

The dislocation can be introduced through a Volterra cut, plastic distortion, eigenstrain, or anisotropic Green-function/Stroh representation [10, 11]. The required output is

$$\boxed{\varepsilon^L(\mathbf{x}) = \varepsilon^L(\mathbf{x}; R_L, \mathbf{b}_L, \mathbf{n}_L, \mathbf{C})}. \quad (14)$$

Practical routes are: (i) anisotropic line-segment superposition, (ii) a DDD field evaluator, (iii) an eigenstrain/FEM elasticity solve, or (iv) an isotropic circular-loop benchmark. A non-singular core regularization is recommended when atomistic core corrections are not supplied explicitly.

4 Point-defect energetics

4.1 Equilibrium elastic dipole

For point-defect species $\alpha \in \{\text{i}, \text{v}\}$, define

$$P_{ij}^{g,\alpha} = -\frac{\partial E_f^\alpha}{\partial \varepsilon_{ij}}. \quad (15)$$

To first order in strain,

$$\boxed{E_g^{\alpha L}(\mathbf{x}) = -P_{ij}^{g,\alpha} \varepsilon_{ij}^L(\mathbf{x})}. \quad (16)$$

For an isotropic dilatation center, $P_{ij}^{g,\alpha} = P_g^\alpha \delta_{ij}$ and the interaction reduces to hydrostatic coupling.

4.2 Saddle-point elastic dipole

For a thermally activated jump in direction $\hat{\mathbf{h}}_k$, define the saddle dipole

$$\mathbf{P}_k^{s,\alpha} = \mathbf{P}^{s,\alpha}(\hat{\mathbf{h}}_k), \quad (17)$$

and

$$\boxed{E_s^{\alpha L}(\mathbf{x}, \hat{\mathbf{h}}_k) = -P_{ij}^{s,\alpha}(\hat{\mathbf{h}}_k) \varepsilon_{ij}^L(\mathbf{x})}. \quad (18)$$

Woo's 1982 theory emphasizes that anisotropy of $\mathbf{P}^{s,\alpha}$ can break the vacancy-loop/interstitial-loop symmetry even within linear elasticity [2]. Dederichs and Schroeder derived the corresponding anisotropic diffusion in stress fields from a microscopic lattice description [3].

4.3 Local migration barrier

Let $E_{m,0}^{\alpha k}$ be the unstrained barrier for jump k . Then

$$\boxed{E_m^{\alpha L}(\mathbf{x}, \hat{\mathbf{h}}_k) = E_{m,0}^{\alpha k} + E_s^{\alpha L}(\mathbf{x}, \hat{\mathbf{h}}_k) - E_g^{\alpha L}(\mathbf{x})}. \quad (19)$$

This is the central bridge between Woo’s saddle-point theory and an atomistically informed finite-loop calculation.

4.4 External stress

If external or microstructural strain is present, use

$$\boldsymbol{\varepsilon}^{\text{tot}} = \boldsymbol{\varepsilon}^L + \boldsymbol{\varepsilon}^{\text{ext}} \quad (20)$$

in both equilibrium and saddle interactions. This permits local crystal-plasticity stress, grain-boundary stress, neighboring-loop fields, and network-dislocation stress to enter the capture calculation.

5 Atomistic near-core energetics

March-Rico and Wirth calculated point-defect binding-energy maps around interstitial a -loops, vacancy a -loops, and vacancy c -loops in α -Zr using molecular statics [7]. Write such an equilibrium map as

$$E_{g,\text{atom}}^{\alpha L}(\mathbf{x}). \quad (21)$$

A direct test of Woo’s mechanism additionally requires saddle maps

$$E_{s,\text{atom}}^{\alpha L}(\mathbf{x}, \hat{\mathbf{h}}_k), \quad (22)$$

obtained by NEB, dimer, or equivalent saddle-search calculations.

A matched elastic–atomistic representation is

$$E_{g,\text{hyb}} = E_{g,\text{el}} + w(d)(E_{g,\text{atom}} - E_{g,\text{el}}), \quad (23)$$

$$E_{s,\text{hyb}} = E_{s,\text{el}} + w(d)(E_{s,\text{atom}} - E_{s,\text{el}}), \quad (24)$$

with $w \rightarrow 1$ near the core and $w \rightarrow 0$ in the elastic far field. This avoids double counting.

6 Discrete jump formulation

For an allowed jump $\mathbf{h}_k = h_k \hat{\mathbf{h}}_k$, transition-state theory gives

$$\boxed{\Gamma_{\alpha k}^L(\mathbf{x}) = \nu_{\alpha k} \exp[-\beta E_m^{\alpha L}(\mathbf{x}, \hat{\mathbf{h}}_k)]}, \quad \beta = (k_B T)^{-1}. \quad (25)$$

For strict detailed balance, forward and reverse transitions should share the same physical saddle state. The lattice master equation is

$$\frac{\partial C_\alpha(\mathbf{x}, t)}{\partial t} = \sum_k \left[\Gamma_{\alpha, -k}^L(\mathbf{x} + \mathbf{h}_k) C_\alpha(\mathbf{x} + \mathbf{h}_k, t) - \Gamma_{\alpha, k}^L(\mathbf{x}) C_\alpha(\mathbf{x}, t) \right] \quad (26)$$

$$+ G_\alpha - R_\alpha. \quad (27)$$

This equation can be solved directly by lattice kinetic Monte Carlo or homogenized to a continuum transport equation.

7 Continuum drift–diffusion formulation

For a dilute defect population,

$$\mu_\alpha = \mu_\alpha^0 + k_B T \ln \left(\frac{C_\alpha}{C_\alpha^{\text{ref}}} \right) + E_g^{\alpha L}(\mathbf{x}). \quad (28)$$

Using an Onsager mobility tensor $\mathbf{M}_\alpha^L$,

$$\mathbf{J}_\alpha = -\mathbf{M}_\alpha^L C_\alpha \nabla \mu_\alpha. \quad (29)$$

Defining $\mathbf{D}_\alpha^L = k_B T \mathbf{M}_\alpha^L$ gives

$$\boxed{\mathbf{J}_\alpha = -\mathbf{D}_\alpha^L(\mathbf{x}) [\nabla C_\alpha + \beta C_\alpha \nabla E_g^{\alpha L}(\mathbf{x})]}. \quad (30)$$

The local diffusion tensor is generated from the stressed jump network, schematically

$$\boxed{D_{\alpha,ij}^L(\mathbf{x}) = \frac{1}{2} \sum_k \Gamma_{\alpha k}^{\text{eff}}(\mathbf{x}) h_{k,i} h_{k,j}}, \quad (31)$$

with Γ^{eff} formed from forward/reverse rates in a detailed-balance-consistent manner.

Far from the loop, HCP diffusion has the form

$$\mathbf{D}_\alpha^\infty = D_{\alpha,a}(\mathbf{e}_{a1} \otimes \mathbf{e}_{a1} + \mathbf{e}_{a2} \otimes \mathbf{e}_{a2}) + D_{\alpha,c} \mathbf{e}_c \otimes \mathbf{e}_c. \quad (32)$$

Thus the same calculation contains diffusion anisotropy difference (DAD), elastic drift, and saddle-point anisotropy.

If the diffusion time $\tau_D \sim L^2/D_\alpha$ is short relative to loop evolution, the quasi-steady problem is

$$\boxed{\nabla \cdot \{ \mathbf{D}_\alpha^L(\mathbf{x}) [\nabla C_\alpha + \beta C_\alpha \nabla E_g^{\alpha L}] \} = 0}. \quad (33)$$

Four independent problems are solved for (i, a_I) , (v, a_I) , (i, a_V) , and (v, a_V) .

8 Slotboom transformation

Define

$$u_\alpha = C_\alpha \exp(\beta E_g^{\alpha L}). \quad (34)$$

Then

$$\nabla C_\alpha + \beta C_\alpha \nabla E_g^{\alpha L} = \exp(-\beta E_g^{\alpha L}) \nabla u_\alpha, \quad (35)$$

so

$$\boxed{\mathbf{J}_\alpha = -\underbrace{\mathbf{D}_\alpha^L \exp(-\beta E_g^{\alpha L})}_{\mathbf{A}_\alpha(\mathbf{x})} \nabla u_\alpha} \quad (36)$$

and

$$\boxed{\nabla \cdot [\mathbf{A}_\alpha(\mathbf{x}) \nabla u_\alpha] = 0}. \quad (37)$$

This form removes the explicit convection-like drift term. If $\mathbf{D}_\alpha^L$ is symmetric positive definite, the transformed equation is a variable-coefficient elliptic problem and is usually preferable for finite-element solution.

9 Boundary conditions

9.1 Outer boundary

For an isolated loop,

$$C_\alpha = C_\alpha^\infty \quad \text{on } \Gamma_\infty. \quad (38)$$

The boundary must be far enough from the loop that the integrated flux is insensitive to its location. For a periodic loop population, periodic or symmetry conditions may be used, with representative cell volume $V_{\text{cell}} \simeq N_L^{-1}$.

9.2 Perfect sink

The simplest core condition is

$$C_\alpha = C_{\alpha,L}^{\text{eq}} \quad \text{on } \Gamma_c. \quad (39)$$

For a pure sink-strength benchmark, $C_{\alpha,L}^{\text{eq}} \rightarrow 0$.

9.3 Finite core reaction

A more general radiation boundary condition is

$$\boxed{\mathbf{n}_c \cdot \mathbf{J}_\alpha = \kappa_{\alpha L} (C_\alpha - C_{\alpha,L}^{\text{eq}}) \quad \text{on } \Gamma_c}, \quad (40)$$

where $\mathbf{n}_c$ points from the matrix into the sink. This permits partial reflection and avoids forcing all near-core kinetics into an arbitrary capture radius.

9.4 Emission and detailed balance

For loop evolution, $C_{\alpha,L}^{\text{eq}}$ should be obtained from detailed balance using the free-energy change for adding/removing a point defect from the loop. This can include line tension, stacking-fault energy, external stress work, and chemical terms.

10 Flux, capture coefficient, sink strength, and bias

The net absorption rate of species α is

$$\boxed{\Phi_{\alpha L} = \int_{\Gamma_c} \mathbf{J}_\alpha \cdot \mathbf{n}_c \, dA}. \quad (41)$$

Define the single-loop rate constant

$$K_{\alpha L} = \frac{\Phi_{\alpha L}}{C_\alpha^\infty - C_{\alpha,L}^{\text{eq}}}. \quad (42)$$

For one loop in a cell of volume V_{cell} ,

$$k_{\alpha L}^2 = \frac{K_{\alpha L}}{D_\alpha^{\text{ref}} V_{\text{cell}}}. \quad (43)$$

The corresponding mean-field sink loss is

$$\left. \frac{\partial C_\alpha}{\partial t} \right|_L = -D_\alpha^{\text{ref}} k_{\alpha L}^2 (C_\alpha - C_{\alpha,L}^{\text{eq}}). \quad (44)$$

Using Woo's convention,

$$\boxed{B_L = 1 - \frac{k_{vL}^2}{k_{iL}^2}}. \quad (45)$$

March-Rico and Wirth use $B_L^{\text{MR}} = Z_{i,L}/Z_{v,L} - 1$, related by

$$B_L^{\text{Woo}} = \frac{B_L^{\text{MR}}}{1 + B_L^{\text{MR}}}. \quad (46)$$

11 Finite-loop growth law

For an ideal prismatic loop,

$$N_L \simeq \frac{\pi R_L^2 b_n}{\Omega}, \quad b_n = |\mathbf{b}_L \cdot \mathbf{n}_L|. \quad (47)$$

With signed defect content $q_L = s_L N_L$,

$$\dot{q}_L = \Phi_{iL} - \Phi_{vL}, \quad (48)$$

so

$$\boxed{\dot{R}_L = \frac{\Omega}{2\pi R_L b_n s_L} (\Phi_{iL} - \Phi_{vL})}. \quad (49)$$

Thus an interstitial loop grows when SIA influx dominates, while a vacancy loop grows when vacancy influx dominates. Cluster absorption/emission can be added to the right-hand side.

12 Relation to earlier theories

12.1 Geometrical sink

If $E_g = E_s = 0$, then

$$\nabla \cdot (\mathbf{D}_\alpha^\infty \nabla C_\alpha) = 0, \quad (50)$$

and only geometry and bulk diffusion anisotropy remain.

12.2 Classical isotropic elastic bias

If

$$\mathbf{P}^{g,\alpha} = P_g^\alpha \mathbf{I}, \quad \mathbf{P}^{s,\alpha} = P_s^\alpha \mathbf{I}, \quad (51)$$

the interaction is dominated by isotropic/hydrostatic coupling. In the harmonic limit, equal-sized vacancy and interstitial loops have nearly the same intrinsic bias. Woo's 1981 effective-medium calculation is a finite-loop implementation of this class of theory [1].

12.3 Woo 1982 saddle-point anisotropy

Woo allowed $\mathbf{P}^{s,\alpha}$ to be anisotropic and showed that the defect/loop interaction is then generally non-harmonic. Reversing loop character changes the sign of the shape-sensitive term, giving

$$B_{a_I} \neq B_{a_V}. \quad (52)$$

In the infinitesimal-loop, dilute-sink limit, Woo obtained analytic reaction radii and bias expressions in terms of saddle-point dipole eigenvalues [2]. The finite-loop calculation should reduce toward this limit under the same assumptions.

12.4 March-Rico–Wirth atomistic thermal drift

March-Rico and Wirth calculated relaxed defect binding-energy maps around finite Zr loops. Spontaneous capture favored SIAs for all loops, whereas thermal-drift capture produced a same-type tendency correlated with local loop stress [7]. Their later cluster-dynamics work showed that thermal-drift capture radii stabilize vacancy a -loop growth and permit concurrent vacancy/interstitial a -loop evolution [8]. The present framework replaces a single thermal-drift capture radius by an explicit transport solution through the full energy and mobility landscapes.

13 Recommended hierarchy of models

A useful decomposition is:

Model G: Geometry only. $E_g = E_s = 0$; retain HCP bulk diffusion anisotropy.

Model I: Isotropic elasticity. Use isotropic equilibrium and saddle dipoles.

Model S: Woo saddle anisotropy. Use anisotropic $\mathbf{P}^s$ and equilibrium $\mathbf{P}^g$ with continuum loop elasticity.

Model B: Atomistic binding only. Use atomistic $E_g(\mathbf{x})$ with bulk $\mathbf{D}_\alpha^\infty$, but no position-dependent saddle correction.

Model H: Hybrid. Use atomistically corrected E_g and E_s , matched to anisotropic continuum elasticity at long range.

Comparisons among these models isolate geometric, DAD, equilibrium-binding, saddle-point, and near-core contributions to the bias.

14 Weak finite-element formulation

Using the Slotboom variable,

$$\nabla \cdot (\mathbf{A}_\alpha \nabla u_\alpha) = 0, \quad \mathbf{A}_\alpha = \mathbf{D}_\alpha^L e^{-\beta E_g^{\alpha L}}. \quad (53)$$

For test function v vanishing on Dirichlet boundaries,

$$\boxed{\int_{\Omega} \nabla v \cdot \mathbf{A}_\alpha \nabla u_\alpha \, dV - \int_{\partial\Omega_N} v \mathbf{n} \cdot \mathbf{A}_\alpha \nabla u_\alpha \, dA = 0}. \quad (54)$$

For a perfect-sink Dirichlet boundary, u_α is prescribed on Γ_c ; for a Robin core reaction, the boundary term is replaced by the transformed reaction law. If $\mathbf{A}_\alpha$ is symmetric positive definite, the bilinear form is elliptic.

15 Potential numerical solution methods

15.1 Finite elements: recommended primary method

A 3-D unstructured tetrahedral FEM is the most flexible option because it naturally handles a toroidal core, anisotropic $\mathbf{D}(\mathbf{x})$, anisotropic elasticity, Robin conditions, periodic boundaries, and local mesh refinement.

FEniCSx. FEniCSx/DOLFINx is well suited because the weak form can be entered directly in UFL and solved with PETSc-based parallel linear algebra [12, 13].

MOOSE. MOOSE is attractive if the finite-loop transport calculation will later be coupled to mechanics, phase-field evolution, cluster dynamics, or other multiphysics models [14, 15].

15.2 Finite volume

A conservative finite-volume method is attractive when exact local flux conservation is important. For strong drift, exponentially fitted/Scharfetter–Gummel fluxes can be used [16], especially when solving the untransformed equation.

15.3 Boundary elements

Boundary-element methods can be efficient for homogeneous transport coefficients and analytic drift potentials, but are much less attractive when $\mathbf{D}_\alpha^L(\mathbf{x})$ varies strongly or atomistic energy maps are used.

15.4 Kinetic Monte Carlo

Lattice or object kinetic Monte Carlo provides an independent validation of the continuum closure and naturally retains discrete jump directions and saddle barriers [17]. It is, however, expensive for high-precision steady capture rates.

15.5 Direct master-equation solve

A sparse deterministic solve of the lattice master equation can bridge KMC and continuum FEM. It is useful for moderate atomistic/lattice domains because it preserves discrete jump topology without stochastic sampling noise.

16 Mesh and domain design

The domain must resolve a core scale of order angstroms and an outer scale of many nanometers. A practical strategy is:

1. construct a toroidal inner surface at $d = r_c$;
2. apply a highly refined boundary-layer mesh around the torus;
3. grow element size geometrically away from the loop;
4. refine where $|\nabla E_g|/(k_B T)$ or saddle corrections are large;
5. enlarge the outer domain until the integrated flux converges.

Recommended convergence checks are

$$\left| \frac{\Phi(h/2) - \Phi(h)}{\Phi(h/2)} \right| < \epsilon_h, \quad \left| \frac{\Phi(2R_\infty) - \Phi(R_\infty)}{\Phi(2R_\infty)} \right| < \epsilon_R. \quad (55)$$

17 Why the Zr problem is genuinely three dimensional

An isotropic circular loop admits axial symmetry, but a prismatic a -loop in HCP Zr generally does not because: (i) the loop plane contains the c axis, (ii) $D_{\alpha,a} \neq D_{\alpha,c}$, (iii) the HCP elastic tensor is anisotropic, (iv) $\mathbf{P}^{s,\alpha}$ depends on crystallographic jump direction, and (v) tilted or elliptical loops further break symmetry. The high-fidelity model should therefore be 3-D, although mirror symmetries may reduce the domain.

18 Dimensionless measures and conditioning

Useful measures are

$$\chi_g = \max_{\Omega} \frac{|E_g|}{k_B T}, \quad \chi_m = \max_{\Omega,k} \frac{|E_s(\hat{\mathbf{h}}_k) - E_g|}{k_B T}. \quad (56)$$

If $\chi_g \gg 1$, the untransformed drift–diffusion equation is stiff and the Slotboom formulation is preferred. A mesh-scale drift estimate is

$$\text{Pe}_h \sim h\beta|\nabla E_g|, \quad (57)$$

for isotropic D ; large values indicate that standard Galerkin treatment of the untransformed equation may require stabilization or exponential fitting.

19 Verification and validation hierarchy

19.1 Verification tests

Test 1: zero interaction field. Set $E_g = E_s = 0$ and reproduce a known geometrical toroidal sink result.

Test 2: equilibrium detailed balance. In a closed domain with no source,

$$C_{\alpha}^{\text{eq}}(\mathbf{x}) \propto e^{-\beta E_g(\mathbf{x})} \quad (58)$$

must produce $\mathbf{J}_{\alpha} = 0$ to numerical tolerance.

Test 3: isotropic saddle limit. Set $\mathbf{P}^{s,\alpha} = P_s^{\alpha} \mathbf{I}$. The intrinsic difference between equal-sized vacancy and interstitial loops should collapse toward the classical harmonic result.

Test 4: Woo 1982 dilute infinitesimal-loop limit. Use Woo’s assumptions and saddle tensors and verify convergence toward his analytical reaction-radius and bias expressions [2].

Test 5: Woo 1981 finite-loop benchmark. Suppress saddle anisotropy and compare finite-loop sink strength and bias with Woo’s effective-medium results [1].

19.2 Validation against atomistics

Test 6: March-Rico binding maps. Use interpolated atomistic $E_g^{\alpha L}(\mathbf{x})$ with no saddle correction and test whether integrated capture tendencies are consistent with the reported radial/normal hierarchy [7].

Test 7: atomistic jump barriers. At selected positions, compare continuum-predicted $E_m = E_s - E_g$ with direct NEB barriers.

19.3 Validation against microstructure evolution

Insert the resulting $K_{\alpha L}(R, T)$ database into cluster dynamics and compare with vacancy/interstitial loop fractions, loop-size distributions, mean radius versus dose, simultaneous a_I/a_V growth, and spatial ordering. March-Rico, Blondel, and Wirth provide a useful cluster-dynamics benchmark [8].

20 Parameter acquisition for α -Zr

Parameter	Source / calculation
$C_{ijkl}(T)$	Experimental or atomistic HCP elastic constants.
$\mathbf{b}_L, \mathbf{n}_L, R_L$	Experimental loop crystallography or prescribed geometry.
$P_{ij}^{g,\alpha}$	DFT or molecular-statics elastic dipoles for vacancy and SIA equilibrium configurations.
$P_{ij}^{s,\alpha}(\hat{\mathbf{h}})$	DFT/atomistic saddle-point elastic dipoles. Chulkin, Chernov, and Sivak reported equilibrium and saddle-point tensors for Zr and their coupling to dislocation stress fields [9].
$E_{m,0}^{\alpha k}$	Unstrained migration barriers for each crystallographic jump family.
$E_{g,\text{atom}}^{\alpha L}(\mathbf{x})$	Molecular-statics binding-energy maps; March-Rico and Wirth provide a direct finite-loop reference [7].
$E_{s,\text{atom}}^{\alpha L}(\mathbf{x}, \hat{\mathbf{h}})$	NEB/dimer calculations around finite vacancy and interstitial loops; this is the key missing quantity for a direct finite-loop test of Woo’s saddle-point mechanism.
$r_c, \kappa_{\alpha L}$	Core-reaction model; calibrate against atomistic absorption trajectories or establish insensitivity by convergence tests.

21 Recommended computational workflow

1. **Choose loop state:** specify $(R_L, \mathbf{b}_L, \mathbf{n}_L, s_L, T)$.
2. **Generate finite-loop field:** compute $\boldsymbol{\varepsilon}^L(\mathbf{x})$ using anisotropic HCP elasticity.
3. **Construct equilibrium energy:** evaluate $E_g^{\alpha L} = -\mathbf{P}^{g,\alpha} : \boldsymbol{\varepsilon}^L$ or use a matched atomistic map.
4. **Construct saddle energies:** for each jump direction, evaluate $E_s^{\alpha L} = -\mathbf{P}_k^{s,\alpha} : \boldsymbol{\varepsilon}^L$ or use atomistic saddle maps.
5. **Compute local jump rates and mobility:** use transition-state rates and form $\mathbf{D}_\alpha^L(\mathbf{x})$.
6. **Solve transport:** solve the Slotboom equation for (i, a_I) , (v, a_I) , (i, a_V) , and (v, a_V) .
7. **Integrate flux:** compute $\Phi_{\alpha L}$ on the core surface.
8. **Construct sink strengths and biases:** calculate $K_{\alpha L}$, $k_{\alpha L}^2$, and B_L .

9. **Tabulate:** store

$$K_{\alpha L} = K_{\alpha L}(R_L, T, \mathbf{b}_L, \mathbf{n}_L, s_L, \sigma^{\text{ext}}). \quad (59)$$

10. **Couple to microstructure evolution:** interpolate the database in SRCD/cluster dynamics or solve on demand for selected states.

22 Coupling to spatially resolved cluster dynamics

For a loop at $\mathbf{x}_L$, the surrounding SRCD field supplies

$$C_\alpha^\infty \rightarrow C_\alpha(\mathbf{x}_L). \quad (60)$$

The net signed loop-content rate becomes

$$\dot{q}_L = K_{iL}[C_i(\mathbf{x}_L) - C_{i,L}^{\text{eq}}] - K_{vL}[C_v(\mathbf{x}_L) - C_{v,L}^{\text{eq}}]. \quad (61)$$

A local surrogate may be constructed as

$$K_{\alpha L} \approx \hat{K}_{\alpha L}(R_L, T, s_L, \mathbf{n}_L, \mathbf{b}_L, \sigma_{ij}^{\text{loc}}), \quad (62)$$

allowing finite-loop transport physics to enter a large microstructure calculation without solving a 3-D PDE around every loop at every time step.

23 Reduced-order strategies

23.1 Offline/online tabulation

Perform high-fidelity calculations over a sparse design in $(R, T, s_L, \text{orientation}, \sigma^{\text{ext}})$, then construct interpolation tables or a physics-constrained surrogate.

23.2 Axisymmetric benchmark

Before the full HCP calculation, solve a circular loop in isotropic elasticity and isotropic diffusion in axisymmetric (r, z) coordinates. This provides an efficient verification platform.

23.3 Local line approximation

For $R_L \gg r_c$, approximate each loop segment locally as a straight edge dislocation with curvature corrections. This provides a bridge to local line-bias models.

23.4 Multipole far field

Use an exact or DDD field near the loop and a multipole expansion in the far field to accelerate repeated calculations.

24 Recommended implementation sequence

1. 2-D axisymmetric isotropic finite loop, constant D , elastic E_g only.
2. Add species-dependent equilibrium elastic dipoles.
3. Add the Slotboom formulation and verify detailed balance.
4. Add Woo saddle-point anisotropy through $\mathbf{P}^s$.
5. Move to 3-D HCP anisotropic diffusion.
6. Replace isotropic loop elasticity by anisotropic HCP loop fields.
7. Add atomistic E_g near-core corrections.
8. Add NEB-derived E_s corrections.
9. Generate $K_{\alpha L}(R, T)$ and $B_L(R, T)$ databases.
10. Couple the database to SRCD/cluster dynamics.

The first decisive comparison should contain

Woo-isotropic vs. Woo-saddle-anisotropic vs. atomistic/hybrid finite-loop transport.	(63)
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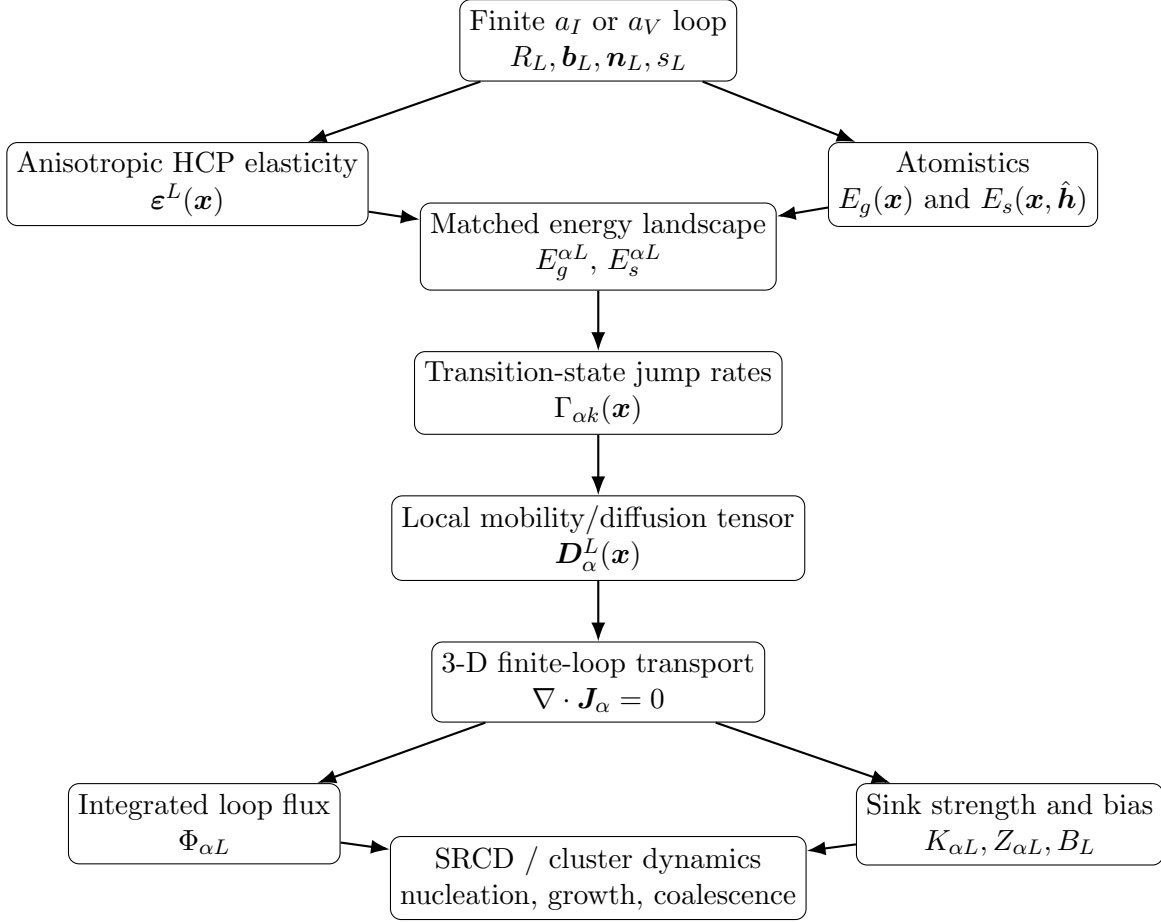
25 Key physical interpretation

The formulation separates two effects that are often conflated:

1. **Thermodynamic binding**, $E_g^{\alpha L}(\mathbf{x})$, which controls local chemical potential and energetic drift.
2. **Kinetic saddle-point modification**, $E_s^{\alpha L}(\mathbf{x}, \hat{\mathbf{h}}) - E_g^{\alpha L}(\mathbf{x})$, which changes local jump rates and the mobility tensor.

March-Rico and Wirth quantitatively characterized the first effect around finite Zr loops through atomistic binding-energy maps and thermal-drift capture criteria [7]. Woo’s 1982 theory emphasized the second effect and showed that saddle-point shape anisotropy can create an intrinsic vacancy-loop/interstitial-loop bias differential [2]. The proposed calculation retains both.

26 Schematic computational architecture



27 Conclusions

A physically complete finite-loop bias calculation for irradiated α -Zr should solve point-defect transport around the actual finite loop rather than assign a fixed empirical bias. The recommended structure is

$$\begin{array}{c}
 \text{finite-loop anisotropic elastic field} \\
 \downarrow \\
 E_g^{\alpha L}(\mathbf{x}), E_s^{\alpha L}(\mathbf{x}, \hat{\mathbf{h}}) \\
 \downarrow \\
 \Gamma_{\alpha k}(\mathbf{x}), \mathbf{D}_\alpha^L(\mathbf{x}) \\
 \downarrow \\
 \nabla \cdot \{ \mathbf{D}_\alpha^L [\nabla C_\alpha + \beta C_\alpha \nabla E_g^{\alpha L}] \} = 0 \\
 \downarrow \\
 \Phi_{\alpha L} \rightarrow K_{\alpha L} \rightarrow Z_{\alpha L} \rightarrow B_L \rightarrow \dot{R}_L.
 \end{array} \tag{64}$$

The formulation contains classical isotropic elastic bias, Woo's saddle-point anisotropy, HCP diffusion anisotropy, and March-Rico-type atomistic loop binding as controlled limits of the same transport problem. The most important new atomistic input required for a decisive comparison is the spatially resolved saddle-point energy landscape around finite vacancy and interstitial a -loops.

References

- [1] C. H. Woo, “The sink strength of a dislocation loop in the effective medium approximation,” *Journal of Nuclear Materials* 98 (1981) 279–294. doi:10.1016/0022-3115(81)90154-9.
- [2] C. H. Woo, “Intrinsic bias differential between vacancy loops and interstitial loops,” *Journal of Nuclear Materials* 107 (1982) 20–30. doi:10.1016/0022-3115(82)90555-4.
- [3] P. H. Dederichs and K. Schroeder, “Anisotropic diffusion in stress fields,” *Physical Review B* 17 (1978) 2524–2536. doi:10.1103/PhysRevB.17.2524.
- [4] K. Schroeder and K. Dettmann, “Diffusion reactions in long range potentials,” *Zeitschrift für Physik B* 22 (1975) 343–350. doi:10.1007/BF01312804.
- [5] A. D. Brailsford, R. Bullough, and M. R. Hayns, “Point defect sink strengths and void-swelling,” *Journal of Nuclear Materials* 60 (1976) 246–256. doi:10.1016/0022-3115(76)90139-2.
- [6] R. Bullough, M. R. Hayns, and C. H. Woo, “The sink strength of dislocation loops and their growth in irradiated materials,” *Journal of Nuclear Materials* 84 (1979) 93–100. doi:10.1016/0022-3115(79)90152-1.
- [7] J. F. March-Rico and B. D. Wirth, “Stress states and point defect capture radii of dislocation *a*-loops and *c*-loops in alpha-zirconium,” *Journal of Nuclear Materials* 587 (2023) 154752. doi:10.1016/j.jnucmat.2023.154752.
- [8] J. F. March-Rico, S. Blondel, and B. D. Wirth, “Atomistically-informed cluster dynamics modeling of concurrent interstitial and vacancy *a*-loop evolution in irradiated alpha-zirconium,” *Acta Materialia* 288 (2025) 120777. doi:10.1016/j.actamat.2025.120777.
- [9] D. A. Chulkin, V. M. Chernov, and A. B. Sivak, “Self-point defects characteristics and their dependence on stress fields of edge and screw basal dislocations with Burgers vector $1/3\langle 11\bar{2}0 \rangle$ in HCP Zr,” *AIP Conference Proceedings* 999 (2008) 146–156. doi:10.1063/1.2918101.
- [10] J. P. Hirth and J. Lothe, *Theory of Dislocations*, 2nd ed., Wiley, New York, 1982.
- [11] T. Mura, *Micromechanics of Defects in Solids*, 2nd ed., Martinus Nijhoff, Dordrecht, 1987.
- [12] I. A. Baratta, J. P. Dean, J. S. Dokken, M. Habera, J. S. Hale, C. N. Richardson, M. E. Rognes, M. W. Scroggs, N. Sime, and G. N. Wells, “DOLFINx: The next generation FEniCS problem solving environment,” 2023. doi:10.5281/zenodo.10447666.
- [13] M. S. Alnæs, A. Logg, K. B. Ølgaard, M. E. Rognes, and G. N. Wells, “Unified Form Language: A domain-specific language for weak formulations of partial differential equations,” *ACM Transactions on Mathematical Software* 40 (2014). doi:10.1145/2566630.
- [14] D. Gaston, C. Newman, G. Hansen, and D. Lebrun-Grandié, “MOOSE: A parallel computational framework for coupled systems of nonlinear equations,” *Nuclear Engineering and Design* 239 (2009) 1768–1778. doi:10.1016/j.nucengdes.2009.05.021.
- [15] C. J. Permann, D. R. Gaston, D. Andrš, R. W. Carlsen, F. Kong, A. D. Lindsay, J. M. Miller, J. W. Peterson, A. E. Slaughter, R. H. Stogner, and R. C. Martineau, “MOOSE: Enabling massively parallel multiphysics simulation,” *SoftwareX* 11 (2020) 100430. doi:10.1016/j.softx.2020.100430.

- [16] D. L. Scharfetter and H. K. Gummel, “Large-signal analysis of a silicon Read diode oscillator,” *IEEE Transactions on Electron Devices* 16 (1969) 64–77. doi:10.1109/T-ED.1969.16566.
- [17] L. Malerba, C. S. Becquart, and C. Domain, “Object kinetic Monte Carlo study of sink strengths,” *Journal of Nuclear Materials* 360 (2007) 159–169. doi:10.1016/j.jnucmat.2006.10.002.